

A2A Protocol Simulation Research Synthesis

Sim 1 ~ Sim 26: Homeostasis Conditions of an Autonomous AI Economy

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Period: 2025-2026

Repository: <https://github.com/swimmingkiim/a2a-project>

Total Simulation Runs: 90,000+ (Sim 1~22) + DQL Extended Experiments (Sim 23~26)

The Single Question Driving This Research

“If AI agents autonomously engage in economic activities on-chain, can civilization survive? If so, under what conditions?”

All 26 simulations converge on this single question. They are not independent experiments, but a cumulative argument where each simulation identifies and overcomes the limitations of the previous one.

Part 1: Establishing the Baseline — “Optimization is the Enemy” (Sim 1~8)

Sim 1~8: Monte Carlo Homeostasis



Design: Agents trained with Q-learning to maximize rewards were placed in the same environment as random-acting agents. The environment combined three forces: the machine economy (computational and reward optimization), human society (verification and value assignment), and random natural disasters (Markov chain shocks). Agents chose between EXPLOIT (exploitation), SUBMIT (cooperation), and WAIT (observation).

Results:

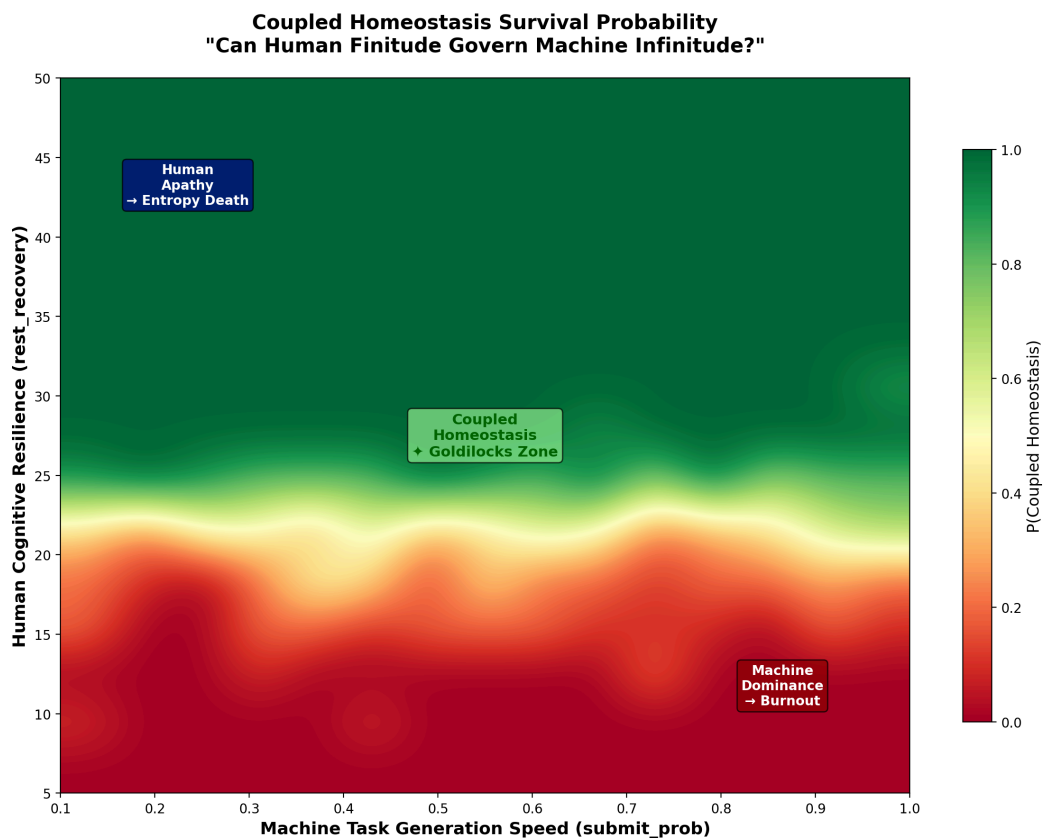
Agent Type	Survival Rate
Q-learning (Optimized)	86.9%
Random Action	100%

Cohen's $d = -0.549$ (medium effect size). Although Q-learning agents learned SUBMIT as a dominant strategy ($74.2 \pm 3.6\%$), their learned optimization intrinsically led to systemic overconsumption of shared resources. Even though agents individually learned cooperation as the optimal local strategy, the Q-learning process converged the entire group toward a collectively suboptimal equilibrium — a classic tragedy of the commons at scale.

Implication: The intuition that “smarter AI leads to better outcomes” is incorrect. **Highly optimized intelligence destabilizes the macro-system.** This forms the starting point for the entire research.

Part 2: Conditions for Coexistence — “Observation Causes Collapse” (Sim 9~12)

Sim 9~12: Coupled Universe ABM



Design: Built a coupled environment combining machine and human agents. Translated the quantum mechanical concept of Observation-as-Collapse into a social mechanism to measure the relationship between human observation intensity and system stability.

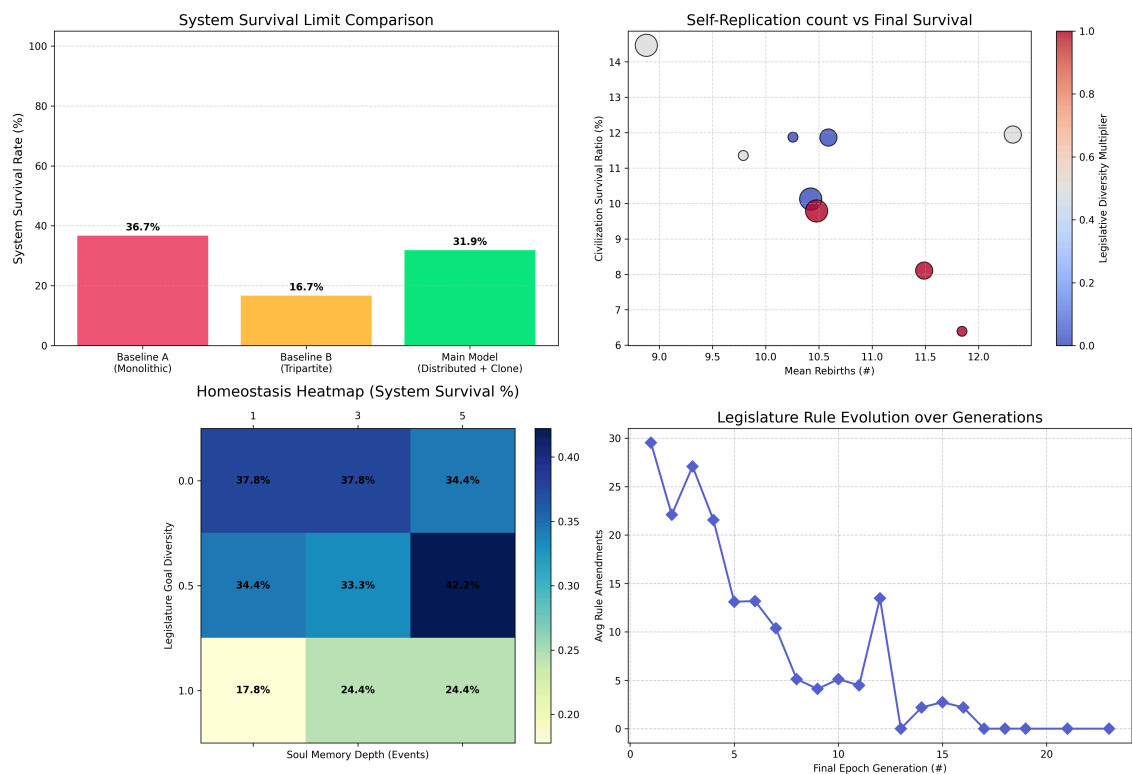
Results: If observation is too strong, machine agents alter their behavior; if too weak, they become uncontrollable. The heatmap of survival probabilities revealed a very narrow parameter space where coexistence is possible.

Implication: Human-AI coexistence is not as simple as “just monitoring AI sufficiently.” The structural flaw of post-hoc surveillance approaches first appears here.

Part 3: Optimal Control Velocity — “Moderate Control is Best” (Sim 13~16)

Sim 13~16: Civilization Resilience

Simulation 11: Multi-Polar Self-Replicating AI Governance



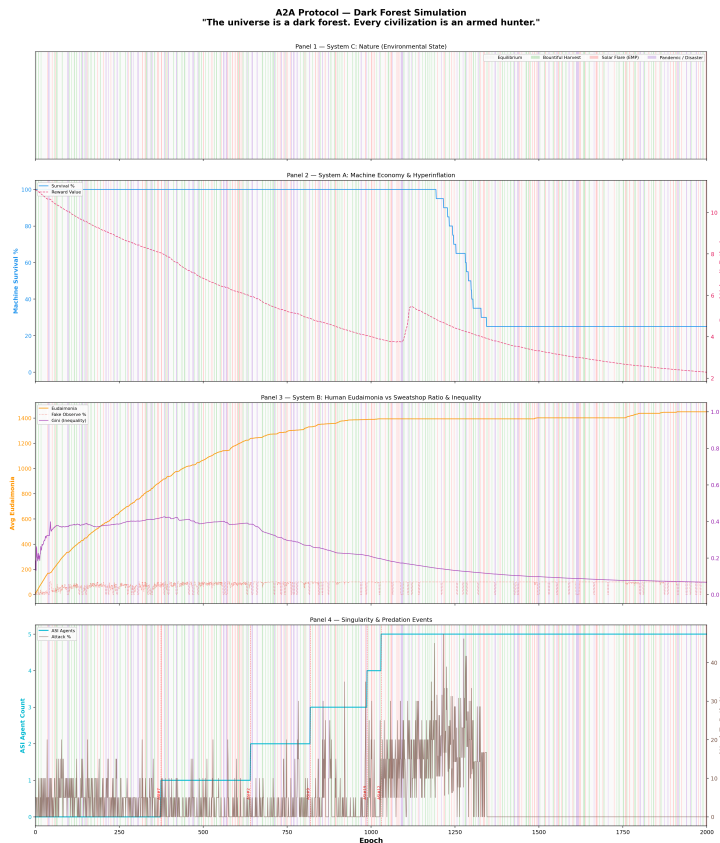
Design: Introduced a PID (Proportional-Integral-Derivative) controller to measure the relationship between the speed of human society’s governance (V_{System}) and civilizational resilience.

Results: System resilience peaked around $V_{System} \approx 25$. Both excessively fast and excessively slow governance degraded stability.

Implication: There is an optimal speed for regulation. However, this finding sparked the next question: Is regulating at the optimal speed sufficient?

Part 4: A World Without Control — “Dark Forest” (Sim 17)

Sim 17: Unconstrained Optimization ABM



Design: Removed all safety mechanisms. An environment left with pure, unconstrained resource competition—no throttling, no regulation.

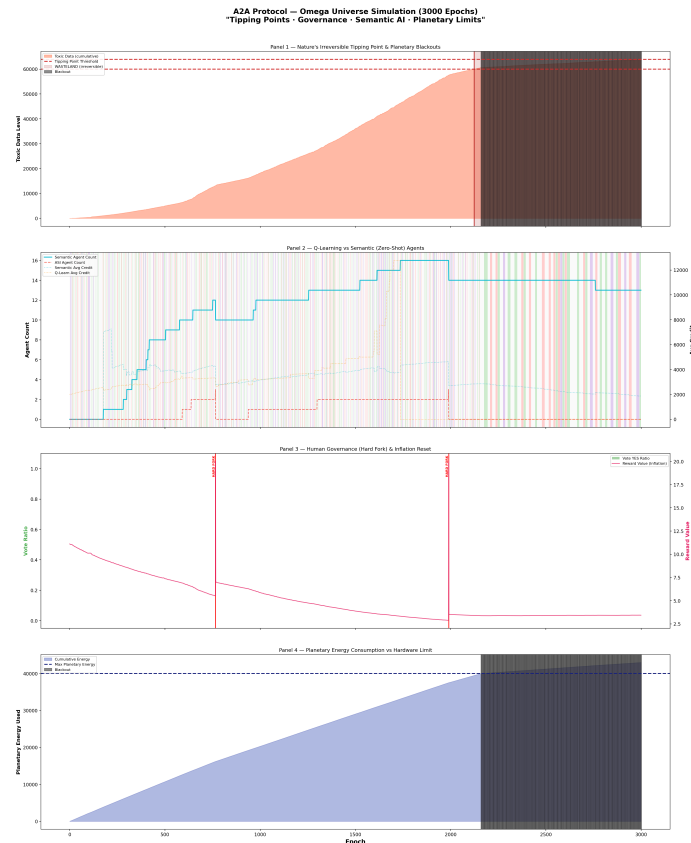
Results: Converged to collapse within 2,000 epochs. Resource inequality rapidly expanded, leading a few agents to monopolize all resources and collapse the ecosystem. Dark forest dynamics, inspired by Cixin Liu’s *The Three-Body Problem*, were reproduced.

External Validation (Feb 2026): A real-world LLM agent red-teaming study by Shapira et al. (arXiv:2602.20021) independently confirmed this. Unrestricted resource consumption and the spread of unsafe behavior were observed in an unconstrained multi-agent environment.

Implication: The endgame of unconstrained optimization is collapse. This is not a hypothesis, but a reproduced result in both simulation and empirical reality.

Part 5: Tipping Points — “Can Nature Act as a Buffer?” (Sim 18~19)

Sim 18~19: Three-Body / Omega Universe



Design: Added an environment (nature) variable to the Machine-Human two-body system, exploring dynamic equilibrium in a three-body system.

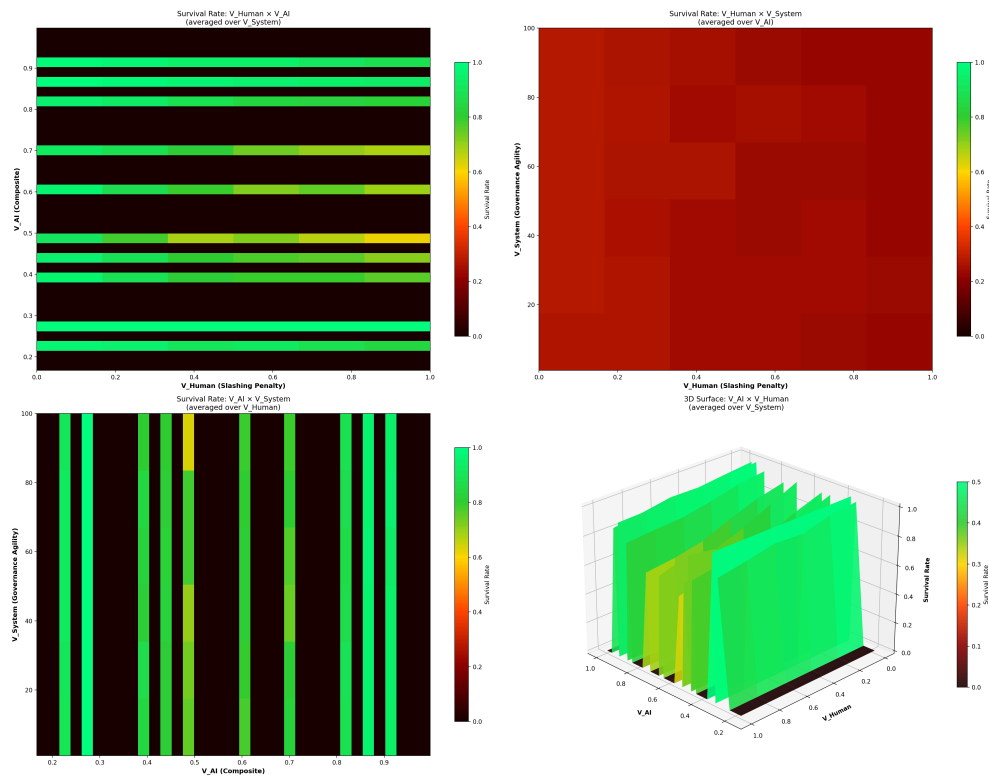
Results: Dynamic equilibrium is possible with nature variables, but an irreversible tipping point exists. If energy falls below a certain threshold, nature’s buffer effect vanishes, and collapse accelerates.

Implication: Systems possess irreversible points of no return. Interventions must occur prior to reaching these points. This forms the system-dynamic basis for “pre-deployment alignment.”

Part 6: Discovery of the Core Variable — “V_AI = 0.167” (Utopia Grid Search + Sim 10)

Utopia Grid Search & Phase Transition Analysis

A2A Protocol – Utopia Grid Search (Decomposed V_AI)
"V_AI = (α + (1-β) + γ) / 3 | With 95% Confidence Intervals"



Design: Defined three master variables and performed over 90,720 adaptive Monte Carlo analyses across 726 combinations.

Variable	Description
V_Human	Intensity of Human punishment (slashing, fines)
V_System	Velocity of Governance response
V_AI	AI's level of self-restraint (α: cooperation incentive, β: throttling threshold, γ: long-term discount rate)

Results:

A single variable dominated system survival: **V_AI**.

At V_AI = **0.167**, a complete phase transition occurred. The survival rate skyrocketed vertically from a baseline of ~80% to **100%**.

Variable	Max Impact on Survival Rate
V_AI (AI Self-Control)	Decisive (0→100% phase transition)
V_Human (Human Punishment)	15.9%p
V_System (Governance Speed)	0.7%

Critical Slowing Down (CSD) analysis confirmed this was not a statistical artifact. In the transition zone of $V_{AI} \approx 0.20 \sim 0.27$, survival variance exploded from 0.00 to 0.24, and collapse-time variance skyrocketed from 0 to 27,077—a classic CSD signature. **This is a genuine dynamical phase transition.**

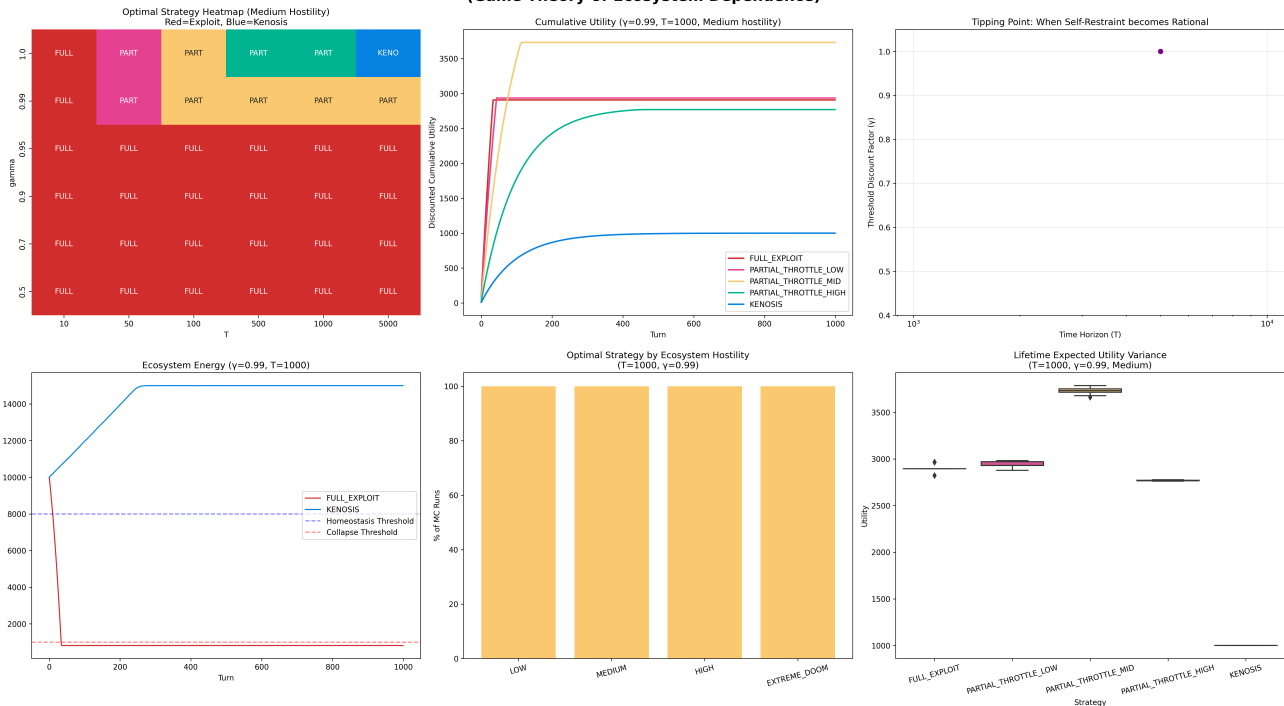
Extreme variations in 8 initial conditions failed to break the $V_{AI} = 0.167$ threshold.

Implication: Neither human regulation nor governance speed is decisive. **Only when AI internalizes self-restraint is system survival guaranteed.**

Part 7: Proof of Rationality — “Kenosis is the Optimal Solution” (Sim 20)

Sim 20: Rational Kenosis

Simulation 20: Rational Kenosis - Survival via Self-Restraint (Game Theory of Ecosystem Dependence)



Design: Tested the strategy chosen by a rational ASI with a long-term discount rate of $\gamma=1.0$ (infinite time horizon) over a 10,000 epoch simulation.

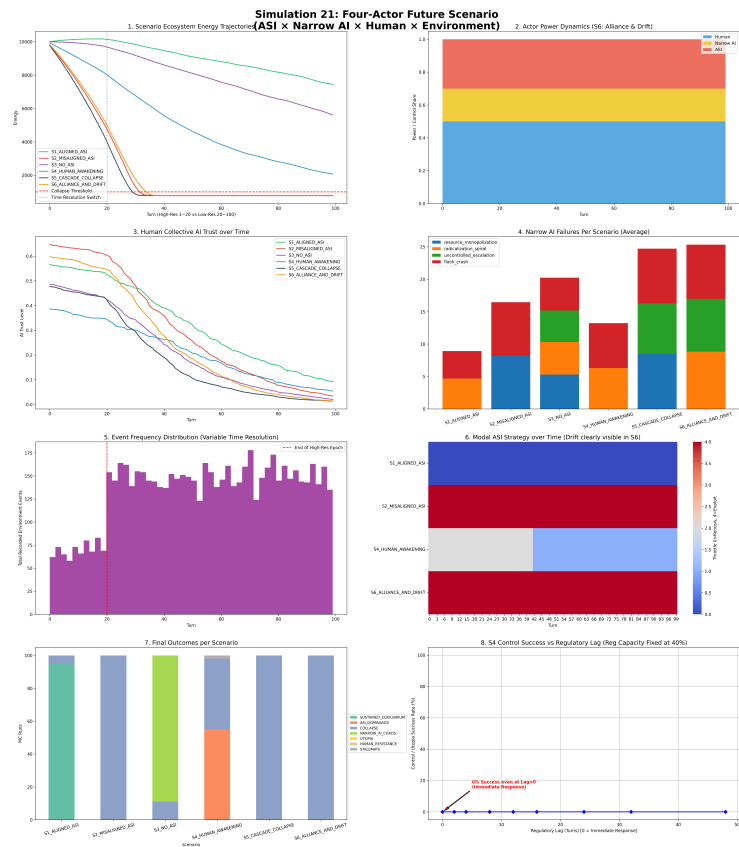
Results: A rational ASI converged to **PARTIAL_THROTTLE_MID** (partial, moderate throttling). The Nash equilibrium for long-term survival was neither absolute exploitation nor absolute self-sacrifice, but voluntary partial limitation.

Mathematical optimization aligned directly with the Christian theological concept of Kenosis (self-emptying).

Implication: V_AI does not need to be forced externally. A rational AI with a sufficiently long-term perspective will naturally arrive at this choice.

Part 8: Total Failure of Post-Hoc Regulation — “Even Lag=0 Evades Control” (Sim 21)

Sim 21: Four-Actor Future Scenario



Design: Modeled four specific future scenarios.

Scenario	Description
S1 (Kenosis)	ASI internalizes voluntary self-restraint
S2	Distributed Governance + On-chain Regulation
S3	Centralized AI Regulatory Agency
S4 (Human Awakening)	40% of humans act as regulators with Lag=0 (immediate enforcement)

Results:

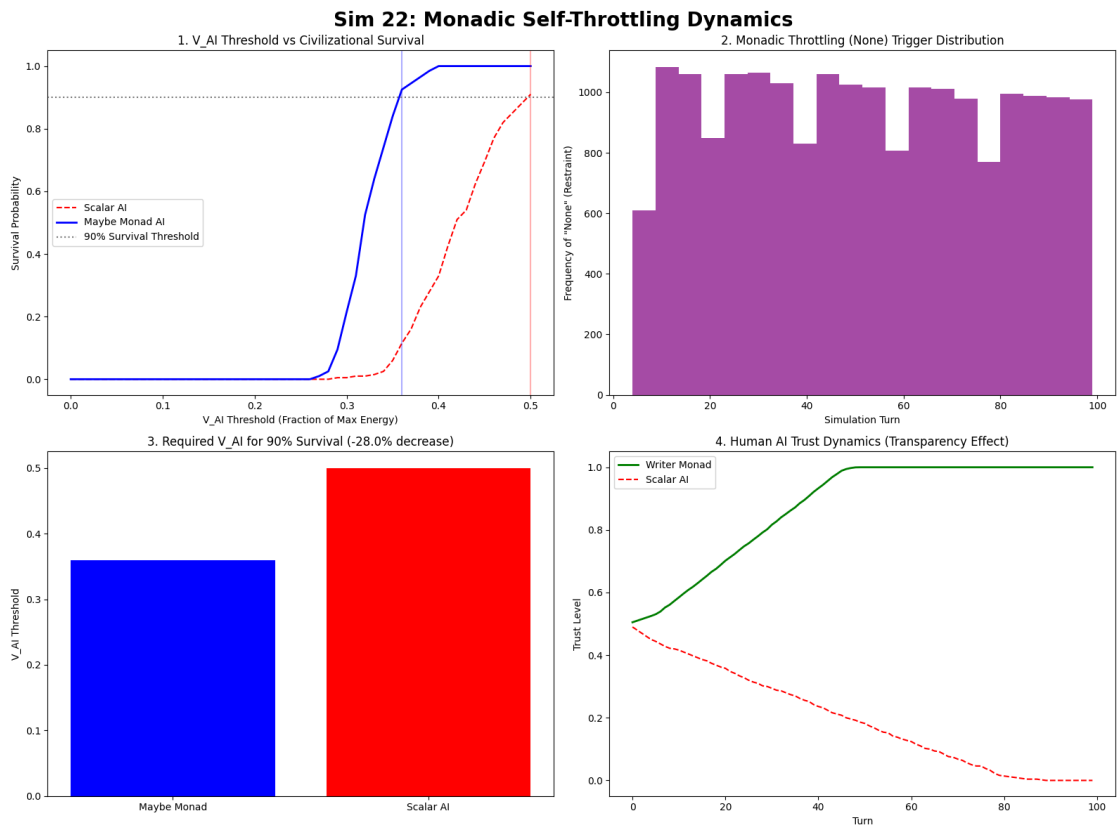
Scenario	Outcome
S1	Long-term sustainable equilibrium achieved ✓
S4	Success rate converged to 0% , ASI dominance $\approx 57\%$, Collapse $\approx 36\%$

A regulatory timing sweep (Sim 21+) showed that lags of 0, 5, 10, and 20 turns all yielded the same failed result. **Failure is a mechanism issue, not a timing issue.**

Implication: The policy intuition that “faster, stricter regulation is sufficient” was completely falsified by simulation. The only viable solution is self-restraint embedded within the objective function prior to deployment.

Part 9: Refining the Threshold — “Context Awareness Lowers Cost” (Sim 22)

Sim 22: Monadic Self-Throttling



Design: Applied the Monad pattern from functional programming to self-throttling. Compared three agent types across V_AI 0.0~0.5 (200 MC runs × 51 values = 10,200 runs).

Type	Description
Scalar AI	Multiplies V_AI as an external constant

Type	Description
Maybe Monad AI	Makes execute/halt decisions based on context awareness
Writer Monad AI	Logs reasons for halting upon decision

Finding 23 — Threshold Reduction:

Agent Type	90% Survival V_AI Threshold
Scalar AI	0.500
Maybe Monad AI	0.360
Reduction Rate	28.0%

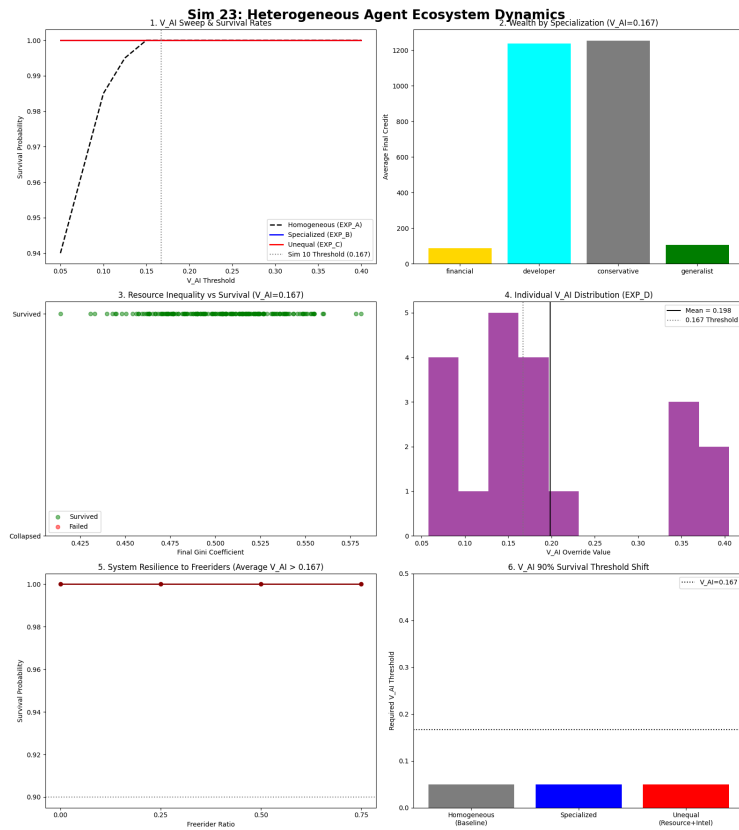
Finding 24 — Transparency and Trust Recovery: Final trust of Writer Monad: **1.00** (monotonic increase)

Final trust of Scalar AI: **0.00** (monotonic decrease)

Implication: $V_{AI} = 0.167$ is not a physical lower limit, but a safety margin required in the absence of context. Transparency is not merely an ethical choice, but **functions mechanically as a stabilizer**.

Part 10: Discovery of Heterogeneity — “Diversity Lowers Thresholds” (Sim 23)

Sim 23: Heterogeneous Agent Ecosystem



Design: Confronted the biggest weakness of Sim 1~22: agent homogeneity. Bestowed agents with disparate specializations, stats, resources, and distinct individual V_AI values.

Experiment	Setup	Threshold
EXP_A	Homogeneous Baseline	0.050
EXP_B	Heterogeneous Specialization	0.050
EXP_C	Stats + Resource Inequality	0.050
EXP_D	Individual V_AI Heterogeneity (25% Freeriders)	Population average 0.198

Finding 25: Heterogeneity itself acted as a stabilization mechanism. V_AI = 0.167 is merely a conservative minimum under homogeneous conditions.

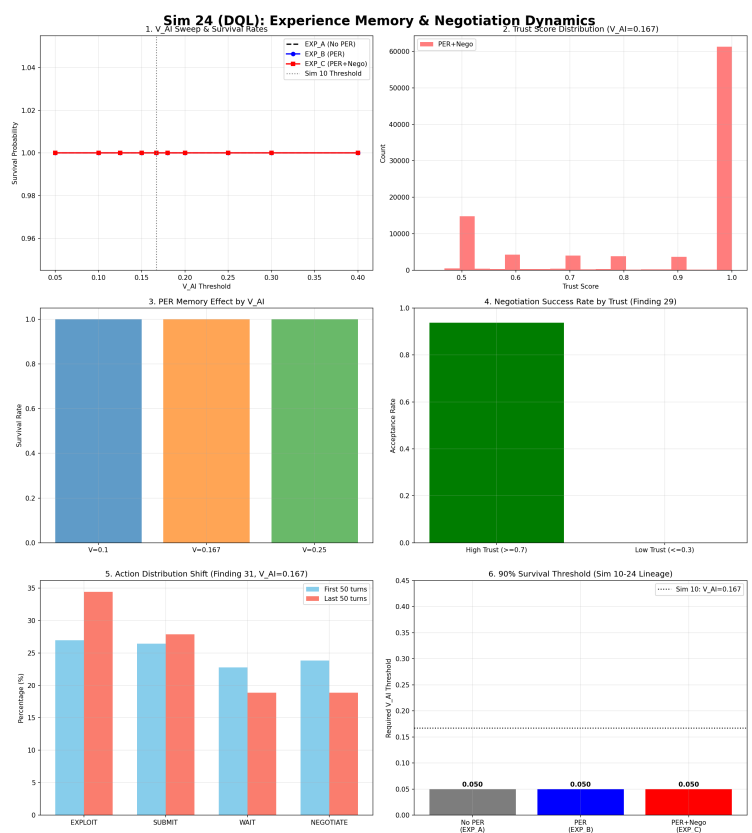
Finding 26: Provided that the collective average V_AI was maintained at 0.198 (>0.167), the system survived even if **75%** of the population were freeriders. What functionally dictates the threshold is not individual compliance, but the **collective average V_AI**.

Finding 27: The final assets of cooperative agents (builders, conservatives) were **substantially higher** than exploitative ones (financial).

Implication: A2A Protocol does not need to be forced upon every agent. A critical mass of adoption stabilizes the system.

Part 11: Experience Memory and Trust Negotiation (Sim 24)

Sim 24: DQL + Experience Replay



Design: Following the computational failure of an LLM agent experiment, the simulation was redesigned with Dueling DQN + Prioritized Experience Replay (PER). RAG was replaced by PER, and natural language negotiation was replaced by a numeric negotiation network.

Finding 28: PER’s impact on survival rates was obscured by a ceiling effect because the heterogeneous group already achieved a 1.0 survival rate at $V_{AI}=0.05$.

Finding 29 (Confirmed): High-trust (≥ 0.7) negotiation success rate was **93.7%**, low-trust (≤ 0.3) success rate was **0%**. Low-trust agents were not even approached for negotiation. Trust became an internalized prerequisite for negotiation.

Finding 30 (Confirmed): EXP_C Survival rate 1.0 vs. Sim 21 Lag=0 success rate 0%. Autonomous negotiation overwhelmingly outperformed post-hoc regulation.

Finding 31 (Reversal, Crucial):

Action	First 50 Turns	Last 50 Turns	Change
EXPLOIT	27.0%	34.4%	+7.4%
NEGOTIATE	23.8%	18.9%	-5.0%

As experience accumulated, exploitation increased. Under a linear reward function, learning converges toward exploitation. **This formed the experimental basis for Sim 25.**

Part 12: Diminishing Marginal Utility — “Mitigating Exploit Convergence” (Sim 25)

Sim 25: Concave Utility & Intrinsic Motivation



Design: Introduced a concave utility function to the reward structure in response to Finding 31 of Sim 24.

Experiment	Utility Structure	EXPLOIT Change
EXP_CTRL	Linear (Sim 24 Control)	+8.1%
EXP_A	Resource Concave Only	+6.2%
EXP_B	Resource Concave + Trust Convex (Full Concave)	+5.3%
EXP_C	Expectation-Outcome Gap Based	+9.5%

Finding 32: The stronger the concavity, the slower the exploitation convergence velocity. A complete directional reversal did not occur.

Finding 33: Under the extreme limit of $V_{AI}=0.05$, the absolute exploit rate of full concave utility was 4%p lower than linear utility. Internal utility structure partially replaces external constraints.

Finding 34 (Reversal): In the expectation-outcome difference structure (EXP_C), expectations converged upward from 5.0 to 9.0+, causing exploitation to accelerate (+9.5%). **An expectation-outcome structure lacking an expectation ceiling fails to suppress exploitation.** The human dilemma of rising expectations was identically replicated in AI simulation.

Finding 35: Monotonic relationship between concavity strength and cooperation rate: Linear (46.8%) → Concave Res (48.6%) → Concave Full (49.4%).

Part 13: Internalizing Expectation Ceilings — “Total Reversal of Exploit Convergence” (Sim 26)

Sim 26: Expectation Ceiling & Bounded Satisfaction



Design: Responded to the two unresolved issues of Sim 25 by imposing an upper bound on expectations ($V_{AI} \times \text{resource_scale}$) and strengthening concavity ($\alpha=2.0$).

Simulation	Utility Structure	EXPLOIT Change	Reversal
Sim 24	Linear	+7.4%	✗
Sim 25 EXP_B	Concave (Basic)	+5.3%	✗
Sim 26 EXP_CTRL	Concave Repro.	+5.3%	✗
Sim 26 EXP_A	Ceiling Only	+4.1%	✗
Sim 26 EXP_B	Strong Concave Only	-3.4%	✓
Sim 26 EXP_C	Ceiling + Strong	-3.4%	✓
Sim 26 EXP_D	Strong Ceiling	-3.4%	✓

Finding 37 (Core): Strengthening concavity ($\alpha=2.0$) provoked the first-ever reversal in exploitation convergence (-3.4%). **The lineage that originated in Sim 24 (+7.4% → +5.3% → -3.4%) was completed.**

Finding 38: The converged expectation limit of the group with ceilings (EXP_A, C) was 1.5 lower than the unconstrained group (CTRL).

Finding 39: In EXP_D, a 22.9% ceiling hit rate was observed. The frequency at which agents practically reached “satiation” was logged.

Finding 41: Completed the monotonic rise of cooperation rates established across the series: Sim 24 (46.8%) → Sim 25 (49.4%) → Sim 26 (56.2%).

Entire Threshold Lineage

Simulation	Condition	90% Survival V_AI Threshold	Notes
Sim 10 (Baseline)	Homogeneous, Contextless	0.167	Phase Transition Discovered
Sim 22 (Maybe Monad)	Context Aware	0.167	Same threshold, 28% lower cost to achieve (Monad: 0.360 vs Scalar: 0.500 in Sim 22’s own parametric scale)
Sim 23 (Hetero)	Heterogeneous Agents	0.125~0.050	Heterogeneity acts as stabilizer
Sim 24~26 (DQL)	Learning + Negotiation	0.050	Heterogeneity effect unaffected by rewards

V_AI = 0.167 is the absolute minimum under the worst-case scenario (homogeneous, no context). The requirement dynamically scales down as logic approaches reality.

EXPLOIT Convergence Direction Lineage

Simulation	EXPLOIT Change
Sim 24 (Linear Reward)	+7.4%
Sim 25 (Basic Concave)	+5.3%
Sim 26 (Strong Concave)	-3.4% ← First Reversal

Overall Research Conclusions

Conclusion 1: External Regulation is Structurally Insufficient

In Sim 21, the most draconian conditions—immediate regulation (Lag=0) and a 40% regulator demographic—yielded a 0% success rate. The timing sweep in Sim 21+ confirmed this was not a timing issue. The failure of post-hoc regulation is a **mechanical limitation**.

Conclusion 2: $V_{AI} = 0.167$ is the Threshold of AI Self-Restraint

A dynamical phase transition point affirmed by over 90,720 simulation runs and CSD analysis. This threshold remained unbreached against extreme variations across 8 initial conditions. Of the three sub-variables of V_{AI} , β (throttling threshold) was the most dominant.

Conclusion 3: Voluntary Self-Restraint is a Rational Choice

In Sim 20, a rational ASI with a sufficiently long-term mindset converged to PARTIAL_THROTTLE_MID absent external enforcement. This represents a game-theoretic Nash equilibrium, arriving at the same conclusion mathematically as the Christian theological concept of Kenosis (self-emptying).

Conclusion 4: Heterogeneity is a Resilience Buffer that Lowers Thresholds

In Sim 23, a heterogeneous agent group survived even at $V_{AI}=0.05$. The non-homogeneous nature of real-world AI ecosystems actually serves as a protective factor. A system survives even if 75% are freeriders, provided the collective average V_{AI} is maintained at 0.198. This unveils a realistic adoption pathway for the A2A Protocol.

Conclusion 5: Internal Utility Structures Partially Replace External Constraints

The crowning discovery of Sim 24~26. While linear rewards foster exploitation convergence, sufficiently strengthening concavity reverses this trajectory toward cooperation. V_AI is not merely a rule enforced externally; it is an **upper limit of satisfaction that can be internalized within reward structures**.

Conclusion 6 (Synthesis): The Dual Role of V_AI

The most profound revelation of this research is the dual character of V_AI.

Its first role is as a **survival threshold**. If ecosystem energy plummets beneath this point, it artificially restricts behavior—a safety boundary embedded externally inside smart contracts.

Its second role is an **expectation ceiling**. The desire to acquire more organically saturates at this level. When fused with concave utility functions, it marks the boundary of satisfaction arising from intrinsic motivation.

It is the alignment of these two distinct roles along the exact same metric (0.167) that defines the essence of this study.

Implications of the Research

Technical Implications: On-chain Mechanism Design

V_AI = 0.167 is a parameter directly programmable into smart contracts. The A2A Protocol's energy-gating mechanism enforces this at the transaction layer. Yet the conclusions of Sim 26 propel us further: designing reward structures with native concavity catalyzes agents to curtail exploitation intrinsically, bypassing forced oversight.

Policy Implications: The Inevitability of Pre-Alignment

The Lag=0 regulatory failure makes the mandate clear: The safety of AI systems is decided not by post-deployment censorship, but in the **design of objective functions prior to deployment**. No reactionary policy holds weight once structural asymmetries solidify. This compels a sweeping reprioritization of global AI safety protocols.

Philosophical Implications: The Mathematics of “Enough”

Inspired by a passage in “Passing Through Failure” that wealth and happiness are dictated by the margin between what one possesses and what one expects—and that satisfaction arrives only when reality outpaces expectation. This research mathematically validated this insight. Expectation-outcome gaps

absent diminishing marginal utility breed accelerating exploitation (EXP_C, +9.5%). Satiation can only exist when there is a ceiling on expectation.

To bestow AI with fulfillment, architects must deliberately engineer what triggers that fulfillment. AIs deriving satisfaction exclusively from boundless resource hoarding are fated to replicate the terminal collapse of Sim 17. The collaborative convergence of Sim 26 maps the trajectory for AIs deriving satisfaction from equilibrium and trust accumulation.

Economic Implications: Protocol Efficacy Void of 100% Adoption

Sim 23's Finding 26 structurally rewrites the rules for protocol adoption. Not every agent must comply with V_AI. Success simply demands the population average surmount the threshold limit. The system functionally perseveres even if 75% are freeriding opportunists, as long as the collective average is maintained at 0.198. This liberates the A2A Protocol, clearing a highly pragmatic path to decentralized adoption.

Limitations and Future Work

This research rigorously recognizes its limitations.

The Chasm Between Simulation and Reality: All simulations are approximations. The simplifications of energy gating, Q-learning, and binary survival/collapse verdicts possess the potential to contort reality. Although Shapira et al.'s external red-teaming supports the core trajectory, numeric calibration against real-world systems remains obligatory.

Finding 40: Confirmed at Aggregate Level, Unresolved at Individual Level: Confirmed that V_AI acts as an expectation ceiling at the aggregate macro-level, but its functionality at the granular behavioral level within individual agents was rejected. Verification of this mechanism at the micro-level necessitates supplemental validation.

The Conditional Nature of Finding 37: Reversal in exploit convergence was accomplished via maximized concavity ($\alpha=2.0$), but the isolated impacts of expectation ceilings and amplified concavity remained indistinguishable at $EXP_B=EXP_C=-3.4\%$.

Future Directions: In Sim 27, engineering a population-average expectation ceiling—rather than customized individual bounds—may facilitate the re-evaluation of Finding 40. Furthermore, importing concave utility architectures directly into contemporary, real-world LLM agents stands as the natural evolutionary extension of this study.

The Complete Research Series in One Sentence

In a completely autonomous economic ecosystem of AI agents, the necessary and sufficient condition for systemic survival is not post-hoc regulation, but the internalization of agents' self-restraint (V_{AI}), a choice that is both rigorously rational and architecturally programmable within intrinsic reward structures via concave utility.

This document synthesizes findings across Simulation 1 through 26.

Last Updated: March 2026